

Entanglement.

$$|\psi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Assume $|\psi\rangle_A = c_0^A |0\rangle + c_1^A |1\rangle$, $|\psi\rangle_B = c_0^B |0\rangle + c_1^B |1\rangle$

$$|\psi\rangle_{AB} = \underbrace{c_0^A c_0^B}_{\frac{1}{\sqrt{2}}} |00\rangle + \underbrace{c_0^A c_1^B}_0 |01\rangle + \underbrace{c_1^A c_0^B}_0 |10\rangle + \underbrace{c_1^A c_1^B}_{\frac{1}{\sqrt{2}}} |11\rangle$$

$$|\psi\rangle_{AB} = \sum_{i,j} c_{ij} |i\rangle_A \otimes |j\rangle_B, \quad c_{ij} \neq c_i^A c_j^B$$

$|\psi^+\rangle$ is not a separable state. Entangled.

Von Neumann entropy.

$$S(\rho) \equiv -\text{Tr}(\rho \log \rho) = \left[-\sum_i \lambda_i \log \lambda_i \right]$$

$$[S < 0 : \text{entangled}]$$

$$|\psi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$\rho = \frac{1}{2} (|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|)$$

$$\begin{matrix} & \langle 00| & \langle 11| \\ \begin{matrix} |00\rangle \\ |11\rangle \end{matrix} & \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \end{matrix} \quad \begin{pmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{pmatrix}$$

$$(1-\lambda)^2 - 1 = \lambda^2 - 2\lambda, \quad \lambda = 0, 2$$

$$-(0 \log_2 0 + 2 \log_2 2) = -2$$

$$A_i, B_i \in \{1, -1\}$$

Bell's inequality. $-1 \leq \langle A_i B_j \rangle \leq 1$

A, B are entangled.

$$-2 \leq E \leq 2$$

A measurement A_1, A_2 . $B: B_1, B_2$.

$$E(A_1 B_1 + A_1 B_2 + A_2 B_1 - A_2 B_2) \leq 2 : \text{separable}$$

Assume:

$$A_1 = \sigma_x, A_2 = \sigma_z, B_1 = \frac{1}{\sqrt{2}}(\sigma_x + \sigma_z), B_2 = \frac{1}{\sqrt{2}}(\sigma_x - \sigma_z)$$

$$E(A_1 B_2) = \langle \psi^+ | A_1 B_2 | \psi^+ \rangle$$

$$= \langle \psi^+ | \left\{ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] \right\} | \psi^+ \rangle$$

$$= \frac{1}{\sqrt{2}} \langle \psi^+ | \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} | \psi^+ \rangle \quad \begin{matrix} |0\rangle & |1\rangle \\ |0\rangle & |1\rangle \end{matrix} \begin{pmatrix} \langle 0| & \langle 1| \\ 1 & 1 \end{pmatrix}$$

$$= \frac{1}{2} \langle \psi^+ | (|0\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 0| + |1\rangle\langle 1|) (|0\rangle\langle 0| + |1\rangle\langle 1|) | \psi^+ \rangle$$

$$= \frac{1}{2\sqrt{2}} \cdot (2) = \frac{1}{\sqrt{2}}$$

$$\epsilon = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}}\right) = \frac{4}{\sqrt{2}} = 2\sqrt{2} > 2.$$

$$\rho_{AB} = P|\psi^+\rangle\langle\psi^+| + (1-P)|\psi^-\rangle\langle\psi^-|$$

$$\rho_{AB} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{2P-1}{2} & 0 \\ 0 & \frac{2P-1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\rho_{AB}^{TB} = \begin{pmatrix} 0 & 0 & 0 & \frac{2P-1}{2} \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ \frac{2P-1}{2} & 0 & 0 & 0 \end{pmatrix}$$

$$\det(\rho_{AB}^{TB} - I \otimes \lambda) = 0.$$

$$-\lambda \left(\frac{1}{2} - \lambda\right) \left(\frac{1}{2} - \lambda\right) (-\lambda) = 0, \quad \lambda = 0, 0, \frac{1}{2}, \frac{1}{2}$$

$\exists$ non-pos. eigenvalue : entangled

Separable when $P = \frac{1}{2}$.

$$\rho_W = \begin{pmatrix} \frac{1+2F}{3} & 0 & 0 & \frac{1-4F}{3} \\ 0 & \frac{2-2F}{3} & 0 & 0 \\ 0 & 0 & \frac{2-2F}{3} & 0 \\ \frac{1-4F}{3} & 0 & 0 & \frac{1+2F}{3} \end{pmatrix}$$

$$\rho_W^{TB} = \begin{pmatrix} \frac{1+2F}{3} & 0 & 0 & 0 \\ 0 & \frac{2-2F}{3} & \frac{1-4F}{3} & 0 \\ 0 & \frac{1-4F}{3} & \frac{2-2F}{3} & 0 \\ 0 & 0 & 0 & \frac{1+2F}{3} \end{pmatrix}$$

$$\det(\rho_W^{TB} - I \otimes \lambda) = 0.$$

$$\frac{1+2F}{3}, \det \begin{pmatrix} \frac{2-2F}{3} & \frac{1-4F}{3} \\ \frac{1-4F}{3} & \frac{2-2F}{3} \end{pmatrix} = 0$$

$$\lambda = \frac{1+2F}{3}, \frac{1+2F}{3}, \frac{1+2F}{3}, 1-2F$$

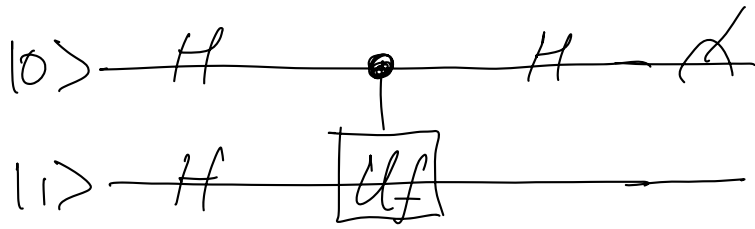
$\exists$ non-pos. eigenvalue: entangled

$$1-2F < 0, \quad -1 \leq 1-F \leq 1$$

$$-1 \leq 1-2F < 0.$$

$$\frac{1}{2} \leq F \leq 1$$

Deutsch



$$|0\rangle|1\rangle \rightarrow (|0\rangle + |1\rangle)(|0\rangle - |1\rangle)$$

$$\rightarrow U_f(|0\rangle + |1\rangle)(|0\rangle - |1\rangle) = U_f(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$

$$\left(\begin{array}{l} U_f: |x\rangle|y\rangle \rightarrow |x\rangle|y \oplus f(x)\rangle \\ |00\rangle = |0, f(0)\rangle, \quad |01\rangle = |0, 1 \oplus f(0)\rangle = |0, \overline{f(0)}\rangle \\ |10\rangle = |1, f(1)\rangle, \quad |11\rangle = |1, 1 \oplus f(1)\rangle = |1, \overline{f(1)}\rangle \end{array} \right)$$

fair: $f(0) \neq f(1)$

$$\Rightarrow \overline{f(0)} = f(1), \quad \overline{f(1)} = f(0).$$

$$\begin{aligned} &\rightarrow |0\rangle|f(0)\rangle - |0\rangle|f(1)\rangle + |1\rangle|f(1)\rangle - |1\rangle|f(0)\rangle \\ &= |-\rangle|f(0)\rangle - |-\rangle|f(1)\rangle = |-\rangle[|f(0)\rangle - |f(1)\rangle]. \end{aligned}$$

$$\boxed{H \rightarrow |1\rangle (\pm |-\rangle)}.$$

biased: $f(0) = f(1)$, $\overline{f(0)} = \overline{f(1)}$.

$$\begin{aligned} &\rightarrow |0\rangle |f(0)\rangle - |0\rangle |\overline{f(0)}\rangle + |1\rangle |f(0)\rangle - |1\rangle |\overline{f(0)}\rangle \\ &= |+\rangle [|f(0)\rangle - |\overline{f(0)}\rangle] \end{aligned}$$

$$\boxed{H \rightarrow |0\rangle (\pm 1-)} \quad \text{}$$

$$\begin{aligned} |0\rangle |1\rangle &\rightarrow |+\rangle |-\rangle \rightarrow \underline{[(-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle] (|0\rangle - |1\rangle)} \\ &\rightarrow \left\{ \underbrace{[(-1)^{f(0)} + (-1)^{f(1)}]}_{\substack{\downarrow \\ f(0) = f(1) \\ \text{Biased}}} |0\rangle + \underbrace{[(-1)^{f(0)} - (-1)^{f(1)}]}_{\substack{\downarrow \\ f(0) \neq f(1) \\ \text{Fair}}} |1\rangle \right\} |-\rangle \end{aligned}$$

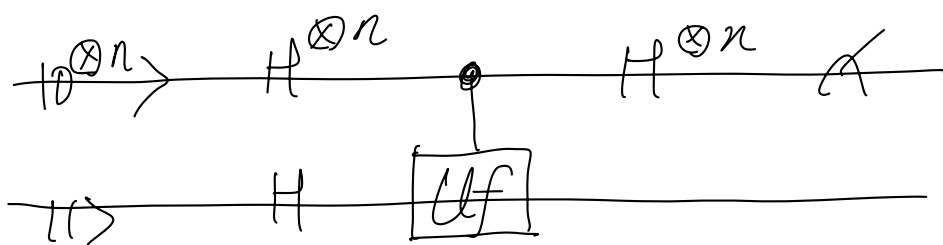
Deutsch-Jozsa

f : constant or balanced?

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \xrightarrow{H} \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

oracle $\rightarrow \frac{1}{\sqrt{2^n}} \left(\sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle \right) \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$

$$H^{(n)} |0\rangle^n = |+\rangle^{\otimes n}$$



$$\begin{aligned} |x\rangle |y\rangle &\rightarrow |x\rangle (y \oplus f(x)) \\ |x\rangle \{ |0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle \} \\ &= (-1)^{f(x)} (|0\rangle - |1\rangle) \end{aligned}$$

$$H \rightarrow \left(\frac{1}{2^n} \sum_{x=0}^{2^n-1} \sum_{y=0}^{2^n-1} (-1)^{f(x)} (-1)^{x \cdot y} |y\rangle \right) \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$\left(H |x\rangle = \prod_{i=1}^n \left(\frac{1}{\sqrt{2}} \sum_{x_i \in \{0,1\}} (-1)^{x_i \cdot y_i} |y_i\rangle \right) = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{x \cdot y} |y\rangle \right)$$

$$x \cdot y = (x_1 \wedge y_1) \oplus (x_2 \wedge y_2) \oplus \dots \oplus (x_n \wedge y_n)$$

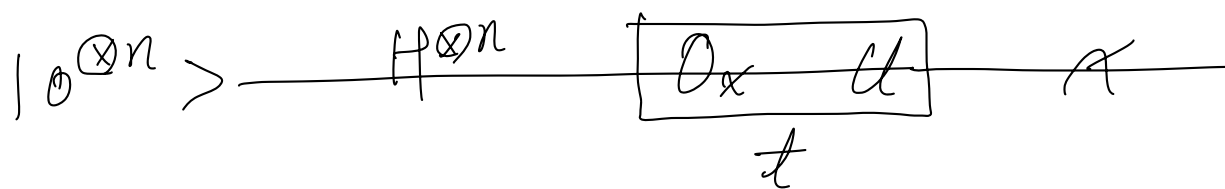
measurement : $\frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x)} (-1)^{x \cdot y}$

constant f : $\frac{1}{2^n} (-1)^{f(0)} \sum_{x=0}^{2^n-1} (-1)^{x \cdot y} = (-1)^{f(0)} \delta_{y,0}$

therefore we measure $(10^n)^n$.

Classical case : $2^{n-1} + 1$ queries.

Grover.



$$O|x\rangle = \begin{cases} |x\rangle & \text{if } x \neq \omega \\ -|x\rangle & \text{if } x = \omega \end{cases} = \begin{bmatrix} & \\ & -1 \end{bmatrix}$$

$$= (-1)^{f(x)} |x\rangle.$$

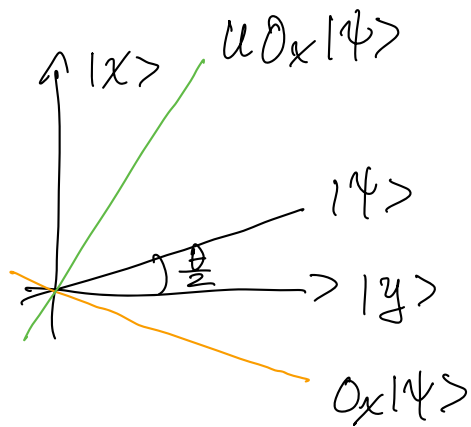
$$O|x\rangle = -|x\rangle \text{ otherwise } O|i\rangle = |i\rangle.$$

$$\begin{aligned} \hat{O}_x &= -|x\rangle\langle x| + \sum_{i \neq x} |i\rangle\langle i| = \sum_i |i\rangle\langle i| - 2|x\rangle\langle x| \\ &= \mathbb{1} - 2|x\rangle\langle x| \end{aligned}$$

$$U = H^{\otimes n} O_{\delta,0} H^{\otimes n}$$

$$O_{\delta,0} = \mathbb{1} - 2|0,0,\dots\rangle\langle 0,0,\dots|$$

$$U = \mathbb{1} - 2|\psi\rangle\langle\psi|.$$



$$|\psi\rangle = \frac{1}{\sqrt{N}} |x\rangle + \frac{\sqrt{N-1}}{\sqrt{N}} |y\rangle$$

$$\cos\left(\frac{\theta}{2}\right) = \frac{\sqrt{N-1}}{\sqrt{N}}$$

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |y\rangle + \sin\left(\frac{\theta}{2}\right) |x\rangle$$

$$\begin{aligned} U O_x |\psi\rangle &= (1 - 2|\psi\rangle\langle\psi|)(1 - 2|x\rangle\langle x|)|\psi\rangle \\ &= (2|\psi\rangle\langle\psi| - 1)(|\psi\rangle - 2\sin\left(\frac{\theta}{2}\right)|x\rangle) \\ &= |\psi\rangle - 4\sin^2\left(\frac{\theta}{2}\right)|\psi\rangle + 2\sin\left(\frac{\theta}{2}\right)|x\rangle \\ &= (1 - 4\sin^2\left(\frac{\theta}{2}\right))|\psi\rangle + 2\sin\left(\frac{\theta}{2}\right)|x\rangle \\ &= \cos\left(\frac{3\theta}{2}\right)|y\rangle + \sin\left(\frac{3\theta}{2}\right)|x\rangle \end{aligned}$$

$$(U O_x)^2 |\psi\rangle = \cos\left(\frac{5\theta}{2}\right)|y\rangle + \sin\left(\frac{5\theta}{2}\right)|x\rangle$$

$$(U O_x)^t |\psi\rangle = \cos\left(\frac{2t+1}{2}\theta\right)|y\rangle + \sin\left(\frac{2t+1}{2}\theta\right)|x\rangle$$

We want $|\psi\rangle \rightarrow |x\rangle$ after t steps.

$$\sin\left(\frac{2t+1}{2}\theta\right) = 1, \quad \left(t + \frac{1}{2}\right)\theta = \frac{\pi}{2}.$$

$\sin\theta \approx \frac{1}{\sqrt{N}}$, because for a large N , $\sin\theta \approx \theta$

$$(2t+1)\frac{1}{\sqrt{N}} = \pi$$

$$t = \frac{1}{2}(\pi\sqrt{N} - 1)$$

$\sqrt{N}$ scale.

2 qubits:

$$\theta = \arcsin \frac{1}{\sqrt{2}} = \frac{\pi}{6}.$$

$$\theta_t = (2t+1)\theta = \frac{\pi}{2}.$$

Example: $n=2$.

$$|\psi\rangle = H^{\otimes n} |0\rangle^{\otimes n} = |+\rangle^{\otimes n} = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

$$(\mathbb{I} - 2|\psi\rangle\langle\psi|)(\mathbb{I} - 2|x\rangle\langle x|)|\psi\rangle \quad \text{add a neg. phase every state orthogonal to } |x\rangle.$$

$$= (\mathbb{I} - 2|\psi\rangle\langle\psi|)|\psi\rangle - 2(\mathbb{I} - |\psi\rangle\langle\psi|)(\langle x|\psi\rangle)|x\rangle$$

$$= (|\psi\rangle - 2|\psi\rangle) - 2\langle x|\psi\rangle|x\rangle + 2|\langle x|\psi\rangle|^2|\psi\rangle$$

$$= (2|\langle x|\psi\rangle|^2 - 1)|\psi\rangle - 2\langle x|\psi\rangle|x\rangle$$

$$= (2\left(\frac{1}{2}\right)^2 - 1)|\psi\rangle - 2\frac{1}{2}|x\rangle$$

$$= -\frac{1}{2}|\psi\rangle - |x\rangle$$

$$= -\frac{1}{2}(|\psi\rangle - |x\rangle) - \frac{3}{2}|x\rangle$$

$$= -\frac{1}{2}[|00\rangle + |01\rangle + |11\rangle + 3|10\rangle] \quad \boxed{\text{if } x=10.}$$

$$\text{Normalization, } \frac{1}{12}[(2|\psi\rangle - |x\rangle) + 3|x\rangle]$$

$$\frac{9}{12} = \frac{3}{4}.$$

$$|\psi\rangle = H^{\otimes 3} |0\rangle^{\otimes 3} = \frac{1}{2\sqrt{2}} \sum_{x=0}^7 |x\rangle$$

$$O_x |\psi\rangle = |\psi\rangle - \frac{2}{2\sqrt{2}} |111\rangle \quad \text{if } x = 111.$$

$$[2|\psi\rangle\langle\psi| - I][|\psi\rangle - \frac{2}{2\sqrt{2}} |111\rangle]$$

$$= 2|\psi\rangle\langle\psi| - |\psi\rangle - \frac{2}{\sqrt{2}} |\psi\rangle\langle\psi|111\rangle + \frac{1}{\sqrt{2}} |111\rangle$$

$$= \frac{1}{2} |\psi\rangle + \frac{1}{\sqrt{2}} |111\rangle = \frac{1}{2} \left[\frac{1}{2\sqrt{2}} \sum_{x=0}^7 |x\rangle \right] + \frac{1}{\sqrt{2}} |111\rangle$$

$$= \frac{1}{4\sqrt{2}} \sum_{x=0}^6 |x\rangle + \frac{1}{4\sqrt{2}} |111\rangle + \frac{1}{\sqrt{2}} |111\rangle$$

$$= \frac{1}{4\sqrt{2}} \sum_{x=0}^6 |x\rangle + \frac{5}{4\sqrt{2}} |111\rangle.$$

Shor.

$$|0\rangle|0\rangle \rightarrow \frac{1}{\sqrt{2^t}} \sum_{k=0}^{2^t-1} |k\rangle|0\rangle$$

$$f(k) = x^k \bmod N.$$

$$\rightarrow \frac{1}{\sqrt{2^t}} \sum_{k=0}^{2^t-1} |k\rangle |x^k \bmod N\rangle.$$

N is given. $1 < k < N$. x is random.

$$7^0 \bmod 15 = 1$$

$$7^1 \bmod 15 = 7$$

$$7^2 \bmod 15 = 4$$

$$7^3 \bmod 15 = 13$$

$$7^4 \bmod 15 = 1.$$

$$\rightarrow \frac{1}{\sqrt{2^t}} [|0\rangle \textcircled{1} + |1\rangle \textcircled{7} + |2\rangle \textcircled{4} + |3\rangle \textcircled{13} + |4\rangle \textcircled{1} + \dots]$$

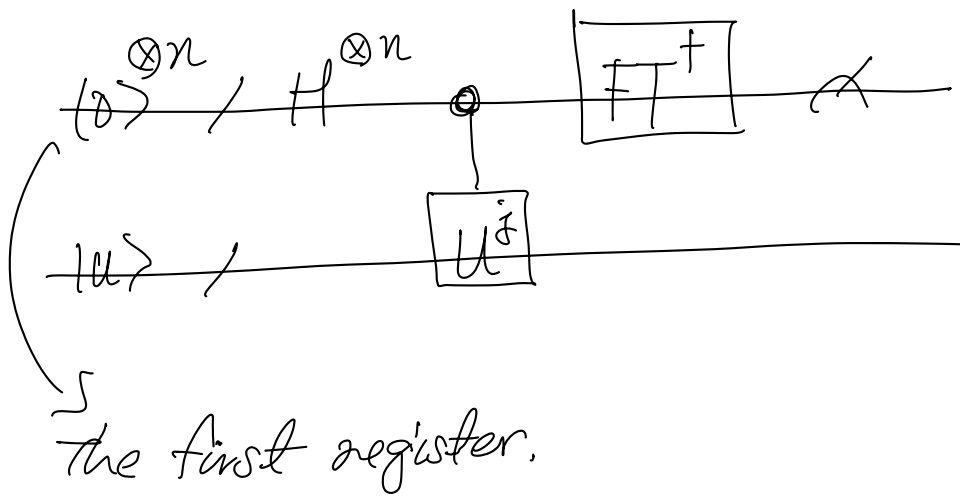
$$\rightarrow \begin{cases} \text{GCD}(7^{\frac{4}{2}} - 1, 15) = 3 \\ \text{GCD}(7^{\frac{4}{2}} + 1, 15) = 5 \end{cases}$$

Quantum F.T.

$$|j\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{2\pi i j k}{N}} |k\rangle$$

Discrete F.T. : $\mathcal{O}(2^n)$

Quantum F.T. : $\mathcal{O}(n^2)$



Shor : n^2
classical : 2^n

$$|0\rangle^{\otimes t} |u\rangle \rightarrow \frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} |j\rangle |u\rangle$$

$$\rightarrow \frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} |j\rangle U^j |u\rangle = \frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} e^{2\pi i j y_n} |j\rangle |u\rangle$$

$\rightarrow$ inverse F.T., $|\tilde{y}_n\rangle |u\rangle$

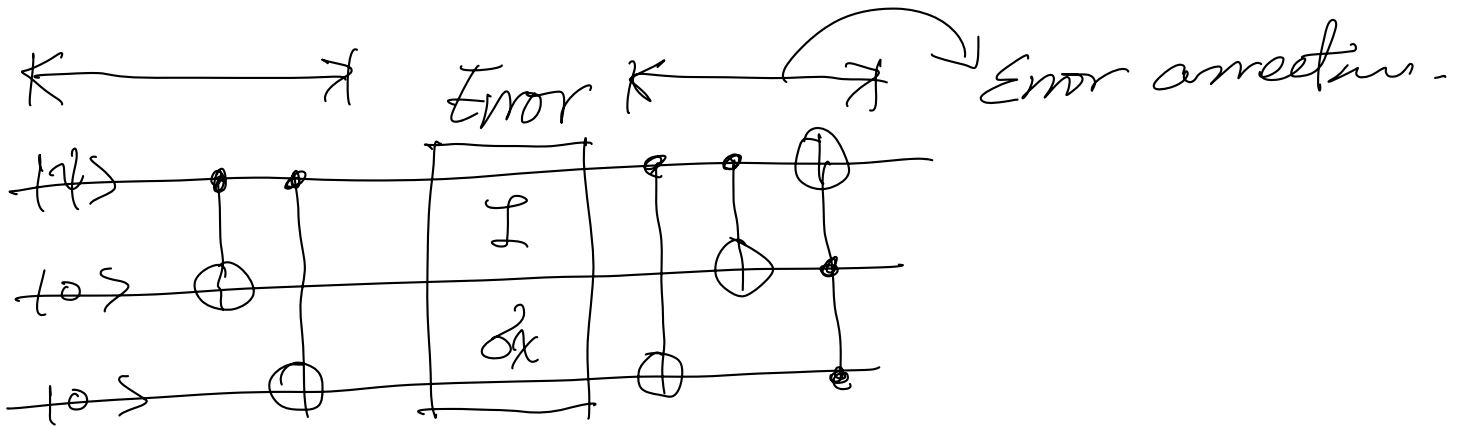
$\rightarrow$ measure the first register, $\frac{1}{\sqrt{2^t}} \sum_{k=0}^{2^t-1} e^{2\pi i y_k} |k\rangle$

$\rightarrow$ measure $\tilde{y}$.

Decoherence. Quantum error correction.

$$\alpha|0\rangle + \beta|1\rangle \rightarrow \alpha|000\rangle + \beta|111\rangle.$$

Encode a superposition state of one qubit with two more qubits.



CNOT gate or Toffoli gate.

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad H \begin{matrix} |0\rangle \\ |1\rangle \end{matrix} = \begin{matrix} |+\rangle \\ |-\rangle \end{matrix}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

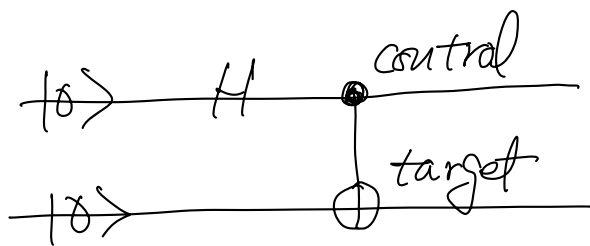
$$H \begin{matrix} |1\rangle \\ |-\rangle \end{matrix}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{Quantum equiv. of NOT gate } \oplus.$$

↑ ↑

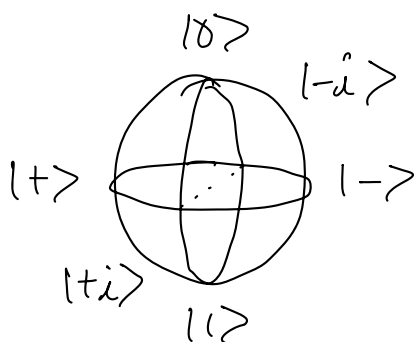
$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_y = i\sigma_x\sigma_z = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$



inverts the target if the control is 1.

$$|00\rangle \rightarrow \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) |0\rangle \rightarrow \frac{1}{2} (|00\rangle + |11\rangle)$$

Bell state.



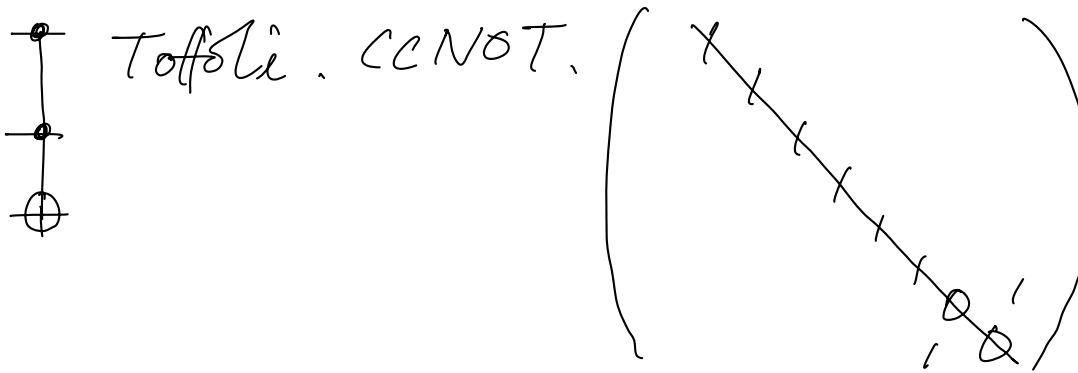
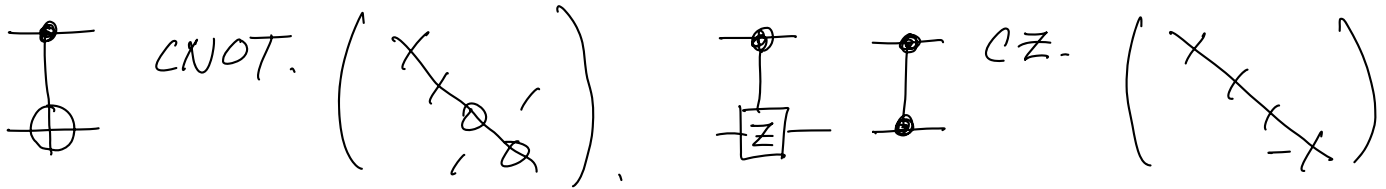
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\rho = |\psi\rangle\langle\psi| = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \begin{pmatrix} \alpha^* & \beta^* \end{pmatrix} = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}$$

$$= \sum_i P_i |\psi_i\rangle\langle\psi_i|.$$

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} : \text{phase.}$$

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{\frac{i\pi}{4}} \end{pmatrix} : \frac{\pi}{8}$$



1. Shannon's entropy. ✓

$$S(p) = - \left(\sum_i p_i \log p_i \right).$$

$$S(\rho) = -\text{Tr}(\rho \log \rho) = - \left(\sum_i \lambda_i \log \lambda_i \right)$$

eigenvalue.
probability.

$$\text{Say } \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right).$$

$$-3 \times \left(\frac{1}{3} \log_3 \frac{1}{3} \right) = 1.$$

$$\frac{\log(\frac{1}{3})}{\log(3)}.$$

$$\log_3 \frac{1}{2} = -\log_3 2.$$

$$\log_3 \left(\frac{1}{2} \right)^2 = -2 \log_3 2$$

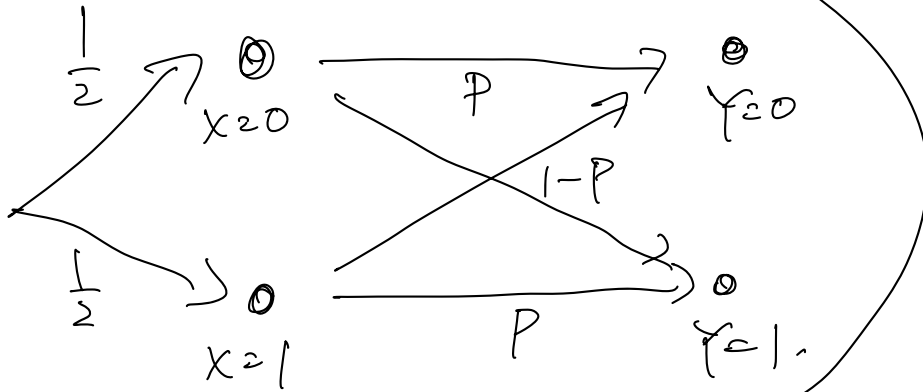
$$\text{For } \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4} \right).$$

$$S = - \left[\frac{1}{2} \log_3 \frac{1}{2} + \frac{1}{4} \log_3 \frac{1}{4} + \frac{1}{4} \log_3 \frac{1}{4} \right].$$

$$\log_a b = \frac{\log b}{\log a}, \quad \log_a b = \frac{1}{\log_b a}$$

$$\log_a \frac{1}{b} = -\log_a b, \quad \log_{\frac{1}{a}} b = -\log_a b$$

2. Channel capacity. ✓

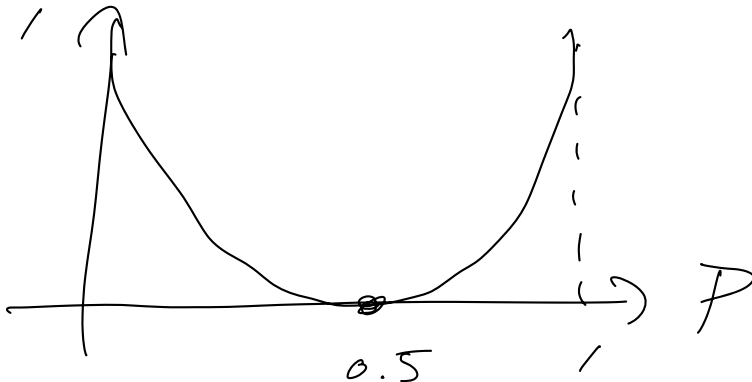


$$R < C = \underbrace{I(X;Y)} = \underbrace{H(X)}_{1} - \underbrace{H(X|Y)}_{\sum_{x=0}^1 P(x) H(X|Y=x)}$$

$$= -P \log P - (1-P) \log (1-P)$$

$$I = 1 + P \log_2 P + (1-P) \log_2 (1-P)$$

Rate of channel utilization.



3. Hadamard op. ✓

$$\hat{H} = |+\rangle\langle 0| + |-\rangle\langle 1|.$$

$$= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\langle 0| + \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\langle 1|.$$

$$= \frac{1}{\sqrt{2}}(|0\rangle\langle 0| + |1\rangle\langle 0| + |0\rangle\langle 1| - |1\rangle\langle 1|)$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

$$\begin{aligned} \hat{H}|\psi\rangle &= \hat{H}(a|0\rangle + b|1\rangle) = a|+\rangle + b|-\rangle \\ &= \left(\frac{a+b}{\sqrt{2}}\right)|0\rangle + \left(\frac{a-b}{\sqrt{2}}\right)|1\rangle = |\psi'\rangle. \end{aligned}$$

$$\begin{cases} |0\rangle, & a=b \\ |1\rangle, & a=-b. \end{cases}$$

4. No cloning theorem. ✓

$$\nexists \hat{U}, \hat{U}|0\rangle = |0\rangle|0\rangle, \hat{U}|1\rangle = |1\rangle|1\rangle.$$

$$\hat{U}|\psi\rangle = |\psi\rangle|\psi\rangle, \quad |\psi\rangle = a|0\rangle + b|1\rangle, \\ \exists a, \exists b, \in \{ |a|^2 + |b|^2 = 1 \}$$

$$(a|0\rangle + b|1\rangle)(a|0\rangle + b|1\rangle) \neq a|00\rangle + b|11\rangle.$$

cannot clone unknown quantum states.

U : unitary operator, is linear

$$U|\psi\rangle = aU|0\rangle + bU|1\rangle \\ \neq U|\psi\rangle = |\psi\rangle|\psi\rangle$$

Entanglement:

$$|\psi\rangle_{AB} = \sum_{i,j} c_{ij} |i\rangle_A \otimes |j\rangle_B, \quad c_{ij} \neq c_i^A c_j^B$$

5. Entanglement: ✓

$|\psi\rangle_{AB} \neq |\psi\rangle_A \otimes |\phi\rangle_B$: entangled.

$$|\psi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle).$$

$$\neq (\alpha|0\rangle + \beta|1\rangle)(\gamma|0\rangle + \delta|1\rangle).$$

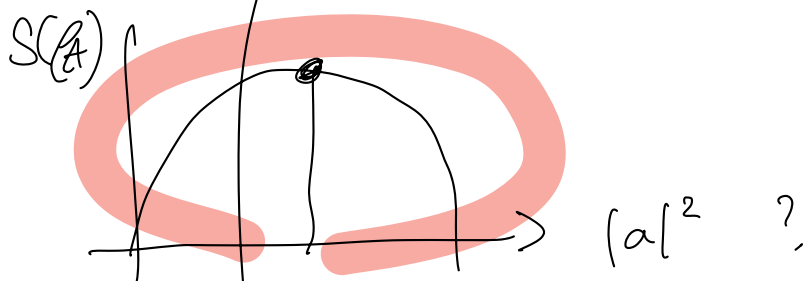
$|\psi\rangle_{AB} = a|00\rangle + b|11\rangle$, maximally entangled.

$$\rho_A = \text{Tr}_B (|\psi\rangle_{AB} \langle \psi|) = \text{Tr}_B (a|00\rangle + b|11\rangle) (\langle 00|a^* + \langle 11|b^*)$$

$$= \text{Tr}_B [|a|^2 |00\rangle \langle 00| + b a^* |11\rangle \langle 00| + a b^* |00\rangle \langle 11| + |b|^2 |11\rangle \langle 11|]$$

$$= |a|^2 |0\rangle \langle 0| + |b|^2 |1\rangle \langle 1| = \begin{pmatrix} |a|^2 & 0 \\ 0 & |b|^2 \end{pmatrix}$$

$$S(\rho_A) = - [|a|^2 \log |a|^2 + |b|^2 \log |b|^2] .$$



$$\text{Tr}_B (|i\rangle\langle j| \otimes |k\rangle\langle l|) = \text{Tr}_B (|i\rangle\langle k| \otimes |j\rangle\langle l|)$$

$$= |i\rangle\langle k| \otimes \text{Tr} (|j\rangle\langle l|) = |i\rangle\langle k| \otimes \langle l|j\rangle$$

$$= \underline{\langle l|j\rangle |i\rangle\langle k|}$$

$$\begin{bmatrix} a_{00} & a_{01} & a_{02} & a_{03} \\ a_{10} & a_{11} & a_{12} & a_{13} \\ a_{20} & a_{21} & a_{22} & a_{23} \\ a_{30} & a_{31} & a_{32} & a_{33} \end{bmatrix} \xrightarrow{\text{Tr}_B} \begin{bmatrix} \text{Tr} \begin{bmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{bmatrix} & \begin{bmatrix} a_{02} & a_{03} \\ a_{12} & a_{13} \end{bmatrix} \\ \begin{bmatrix} a_{20} & a_{21} \\ a_{30} & a_{31} \end{bmatrix} & \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} a_{00} + a_{11} & a_{02} + a_{13} \\ a_{20} + a_{31} & a_{22} + a_{33} \end{bmatrix}$$

6. Schmidt decomposition, (?)

$$|\psi\rangle = \sum_{i=0}^1 c_i |\phi_i \chi_i\rangle$$

$$= c_0 |\phi_0 \chi_0\rangle + c_1 |\phi_1 \chi_1\rangle$$

$$= a |00\rangle + b (|01\rangle + |10\rangle)$$

$$|\phi_0\rangle = \alpha |0\rangle + \beta |1\rangle$$

$$|\chi_0\rangle = \gamma |0\rangle + \delta |1\rangle$$

$$|\phi_1\rangle = \beta |0\rangle - \alpha |1\rangle$$

$$|\chi_1\rangle = \delta |0\rangle - \gamma |1\rangle$$

$$(|\phi_1\rangle = |\phi_0^\perp\rangle)$$

$$|\psi\rangle = c_0 [(\alpha |0\rangle + \beta |1\rangle)(\gamma |0\rangle + \delta |1\rangle)]$$

$$+ c_1 [(\beta |0\rangle - \alpha |1\rangle)(\delta |0\rangle - \gamma |1\rangle)]$$

$$= (c_0 \alpha \gamma + c_1 \beta \delta) |00\rangle \rightarrow a |00\rangle$$

$$+ (c_0 \alpha \delta - c_1 \beta \gamma) |01\rangle \rightarrow b |01\rangle$$

$$+ (c_0 \beta \gamma - c_1 \alpha \delta) |10\rangle \rightarrow b |10\rangle$$

$$+ (c_0 \beta \delta + c_1 \alpha \gamma) |11\rangle \rightarrow 0$$

$$\rightarrow (C_0 + C_1) \alpha \delta - (C_0 + C_1) \beta \gamma = 0$$

$$(C_0 + C_1) (\alpha \delta - \beta \gamma) = 0$$

$$\alpha \delta = \beta \gamma.$$

$$\alpha = \gamma, \delta = \beta$$

$$\rightarrow (C_0 - C_1) (\beta \delta - \alpha \gamma) = -a$$

$$= (C_0 - C_1) (\beta^2 - \alpha^2) = -a$$

$$(C_0 - C_1) = -\frac{a}{(\beta^2 - \alpha^2)}$$

$$\alpha^2 + \beta^2 = 1, \quad C_0^2 + C_1^2 = 1.$$

$$\Rightarrow \boxed{C_0 = \frac{4}{3}a, \quad C_1 = \frac{1}{3}a} \quad (?)$$

$$a = \frac{3}{\sqrt{17}}.$$

$$S(\rho_A), \quad \rho_A = \text{Tr}_B (|\psi\rangle_{AB}\langle\psi|)$$

$$= c_0 |\phi_0\rangle\langle\phi_0| + c_1 |\phi_1\rangle\langle\phi_1|$$

$$S(\rho_A) = - \left(|c_0|^2 \log |c_0|^2 + |c_1|^2 \log |c_1|^2 \right)$$

4. Teleportation of the state $|+\rangle$.
using partially entangled $|\psi\rangle = a|00\rangle + b|11\rangle$.

Average fidelity? $F(|\psi\rangle, |\phi\rangle) = |\langle\phi|\psi\rangle|^2$.
 $0 \leq F \leq 1$.

$$|\psi\rangle_{AA'} |\Phi^{\pm}\rangle_{AB} = (\alpha|0\rangle + \beta|1\rangle)(a|00\rangle + b|11\rangle).$$

$$= |\psi^{\pm}\rangle_{AA'} \frac{1}{\sqrt{2}} (\alpha a|0\rangle + \beta b|1\rangle)_B + |\psi^{\mp}\rangle_{AA'} \frac{1}{\sqrt{2}} (\alpha a|0\rangle - \beta b|1\rangle)_B$$

$$+ |\phi^{\pm}\rangle_{AA'} \frac{1}{\sqrt{2}} (\alpha b|0\rangle + \beta a|1\rangle)_B + |\phi^{\mp}\rangle_{AA'} \frac{1}{\sqrt{2}} (-\alpha b|0\rangle + \beta a|1\rangle)_B$$

$$\left(|\psi^{\pm}\rangle = \frac{1}{\sqrt{2}} (|01\rangle \pm |10\rangle), |\phi^{\pm}\rangle = \frac{1}{\sqrt{2}} (|00\rangle \pm |11\rangle) \right)$$

$$|\langle B\psi^+ | \psi \rangle_{A'}|^2 = \frac{1}{2} (\alpha^2 a + \beta^2 b)^2$$

8. Encoding of Q.T. ✓

$$\begin{array}{l} \frac{1}{2} \nearrow |0\rangle \\ \frac{1}{2} \searrow |1\rangle \end{array}$$

$$\rho = \frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1|$$

$$= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \end{pmatrix} = \begin{pmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \end{pmatrix}$$

$$|\rho - \lambda I| = 0$$

$$\left(\frac{3}{4} - \lambda\right)\left(\frac{1}{4} - \lambda\right) - \frac{1}{4} \cdot \frac{1}{4}$$

$$= \frac{3}{16} - \lambda + \lambda^2 - \frac{1}{16} = \lambda^2 - \lambda + \frac{1}{8}$$

$$\lambda = \frac{1 \pm \sqrt{1 - \frac{1}{2}}}{2} = \frac{1}{2} \left(1 \pm \frac{1}{\sqrt{2}}\right)$$

$$\rho = \begin{pmatrix} \frac{1}{2} \left(1 + \frac{1}{\sqrt{2}}\right) & 0 \\ 0 & \frac{1}{2} \left(1 - \frac{1}{\sqrt{2}}\right) \end{pmatrix}$$

$$\underline{S(\rho) = - \sum_i \lambda_i \log \lambda_i}$$

9. State discrimination - ②

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$\rho = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|)$$

Q. Measure $\{\hat{P}_0, \hat{P}_1\} = \{|0\rangle\langle 0|, |1\rangle\langle 1|\}$

$$\langle\psi|\hat{P}_0|\psi\rangle = P_0 = \left(\frac{1}{2}\right), \quad \langle\psi|\hat{P}_1|\psi\rangle = P_1 = \left(\frac{1}{2}\right)$$

$$\text{Tr}(\hat{P}_0\rho) = P_0 = \left(\frac{1}{2}\right), \quad \text{Tr}(\hat{P}_1\rho) = P_1 = \left(\frac{1}{2}\right)$$

② Measure $\{\hat{P}_+, \hat{P}_-\} = \{|+\rangle\langle +|, |-\rangle\langle -|\}$

$$\langle\psi|\hat{P}_+|\psi\rangle = P_+ = \left(1\right), \quad \langle\psi|\hat{P}_-|\psi\rangle = P_- = \left(0\right)$$

$$\rho = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2} (|+\rangle\langle +| + |-\rangle\langle -|)$$

$$\text{Tr}(\rho\hat{P}_+) = \left(\frac{1}{2}\right) = P_+, \quad \text{Tr}(\rho\hat{P}_-) = \left(\frac{1}{2}\right) = P_-$$